

E-Commerce Sales & Customer Analytics Dashboard

Executive Summary

This project analyzes transactional e-commerce data to uncover revenue trends, customer purchasing behavior, product performance, and geographic sales patterns. Using Python for data preparation and Power BI for visualization, an interactive business dashboard was developed to support data-driven decision-making.

The analysis was performed on 392,692 cleaned transaction records covering the period from December 2010 to December 2011. The business generated a total revenue of £8.89 million from 4,338 customers and 3,877 unique products.

Key findings reveal strong dependence on the United Kingdom market, significant seasonality with peak sales in November 2011, and revenue concentration among a small number of products and customers.

1. Business Objective

The primary objective of this project was to transform raw transactional data into meaningful business insights that could help stakeholders:

- Monitor sales performance.
 - Identify top-performing products and markets.
 - Understand customer purchasing behavior.
 - Discover seasonal sales trends.
 - Support strategic business decisions through interactive reporting.
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2. Dataset Overview

Dataset Source

Online Retail Dataset (Kaggle)

<https://www.kaggle.com/datasets/ulrikthgepedersen/online-retail-dataset>

Real e-commerce transactions with product, quantity, and revenue data.

Dataset Summary

Metric	Value
Date Range	Dec 2010 – Dec 2011
Transactions	392,692
Customers	4,338
Products	3,877
Total Revenue	£8,887,208.89

The dataset contains invoice-level transaction records including product details, quantities, prices, customer identifiers, and country information.

3. Data Cleaning & Preparation

Several preprocessing steps were performed to ensure data quality and reliability.

Data Cleaning Activities

- Removed duplicate records.
- Removed transactions with missing Customer IDs.
- Removed invalid quantity and price values.
- Converted InvoiceDate into datetime format.
- Standardized column formats.

Feature Engineering

New business-focused features were created:

- Revenue = Quantity × Unit Price
- Year
- Month
- Month Number
- Quarter
- Day of Week

These features enabled time-series analysis and dashboard filtering.

4. Key Performance Indicators (KPIs)

The following KPIs were developed and displayed within the Power BI dashboard:

KPI	Value
Total Revenue	£8.89M
Total Orders	18.53K
Total Customers	4.34K
Average Order Value	£479.56

These metrics provide a high-level overview of business performance.

5. Business Analysis & Findings

5.1 Revenue Trend Analysis

Revenue showed an overall upward trend throughout 2011.

The highest revenue month was:

November 2011

Revenue: £1,156,205.61

A significant increase was observed during Q4, indicating strong seasonal demand.

Business Interpretation

The company experiences peak performance during the holiday season. Inventory planning and marketing investments should be increased before Q4 to maximize sales opportunities.

5.2 Geographic Revenue Analysis

Revenue is highly concentrated in a small number of countries.

Top Market

United Kingdom: **£7,285,024.64**

Contribution to Total Revenue: **81.97%**

Other important markets include:

- Netherlands
- EIRE (Ireland)
- Germany
- France

Business Interpretation

The business is heavily dependent on the UK market. While this market remains highly valuable, future growth opportunities may exist through expansion into high-performing European countries.

5.3 Product Performance Analysis

The highest revenue-generating products were:

Product	Revenue
PAPER CRAFT , LITTLE BIRDIE	£168,469.60
REGENCY CAKESTAND 3 TIER	£142,264.75
WHITE HANGING HEART T-LIGHT HOLDER	£100,392.10

Business Interpretation

A relatively small number of products contribute a significant share of revenue. These products should receive priority in inventory planning, promotions, and marketing campaigns.

5.4 Customer Analysis

Revenue is concentrated among a small number of high-value customers.

Top Customers

Customer ID	Revenue
14646	£280,206.02
18102	£259,657.30

Customer ID	Revenue
17450	£194,390.79

Business Interpretation

The company should consider customer segmentation strategies, loyalty programs, and personalized marketing campaigns to retain high-value customers.

5.5 Quarterly Performance Analysis

Quarterly revenue analysis revealed that:

- Q4 generated the highest revenue.
- Sales accelerated significantly toward the end of the year.

Business Interpretation

Seasonality plays a major role in business performance. Demand forecasting should account for year-end spikes to avoid stock shortages and fulfillment issues.

6. Power BI Dashboard

An interactive Power BI dashboard was developed to allow stakeholders to explore business performance dynamically.

Dashboard Components

- Total Revenue KPI
- Total Orders KPI
- Total Customers KPI
- Average Order Value KPI
- Monthly Revenue Trend
- Revenue by Country
- Quarterly Revenue Analysis
- Top Products by Revenue
- Country Filter

- Month Filter

The dashboard enables users to analyze performance across multiple dimensions without requiring technical expertise.

7. Strategic Recommendations

Based on the analysis, the following recommendations are proposed:

1. Prepare for Peak Demand

Increase inventory levels and operational readiness before Q4 to capitalize on seasonal demand.

2. Expand International Presence

Reduce dependency on the UK market by investing in high-potential countries such as Germany, France, Netherlands, and Ireland.

3. Prioritize Top Products

Allocate marketing and inventory resources toward the highest revenue-generating products.

4. Strengthen Customer Retention

Develop VIP and loyalty programs targeting high-value customers to maximize customer lifetime value.

5. Optimize Marketing Timing

Revenue patterns indicate strong purchasing activity during late-week mornings. Marketing campaigns should be tested around these high-performing periods.

8. Power BI Dashboard

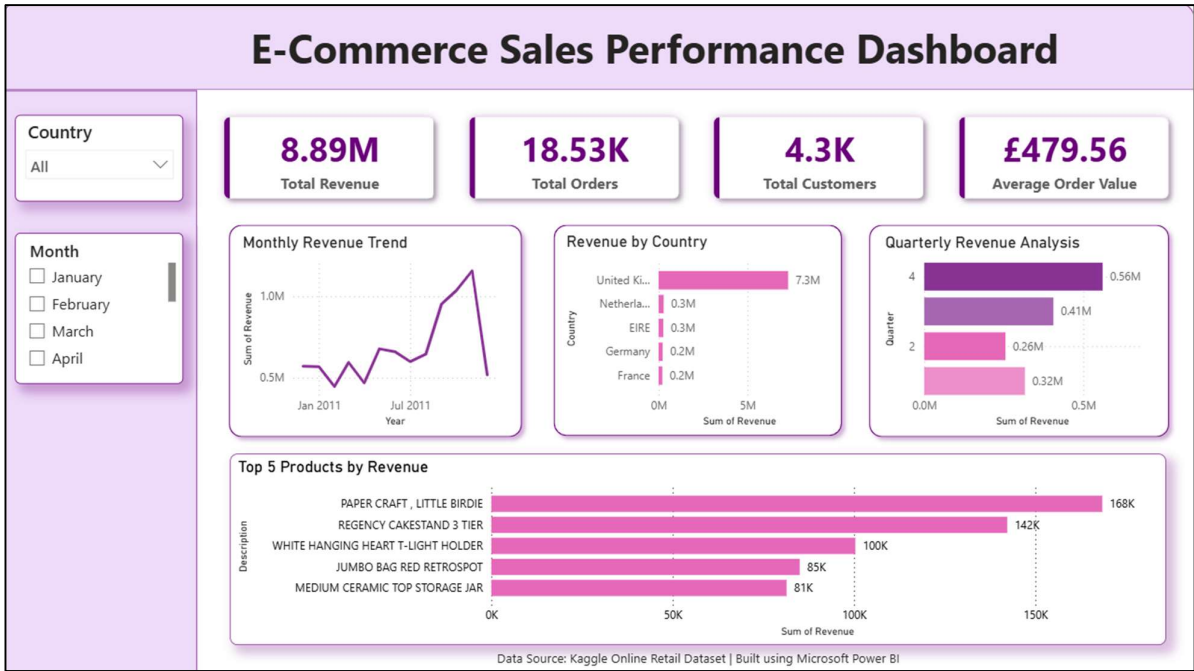


Figure: Interactive Power BI Dashboard showing revenue, customer, order, and product performance metrics.

9. Conclusion

This project demonstrates how data analytics can transform raw transactional data into actionable business intelligence.

Through data cleaning, feature engineering, exploratory analysis, and dashboard development, valuable insights were uncovered regarding revenue trends, customer behavior, product performance, and geographic sales distribution.

The resulting Power BI dashboard provides decision-makers with a practical tool for monitoring business performance and supporting future growth strategies.